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CSE-4355 Electromechanical Systems and Sensors

THE UNIVERSITY OF TEXAS AT ARLINGTON



LAB 4 REPORT

Submitted toward the partial completion of the requirements for CSE 4355

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Introduction

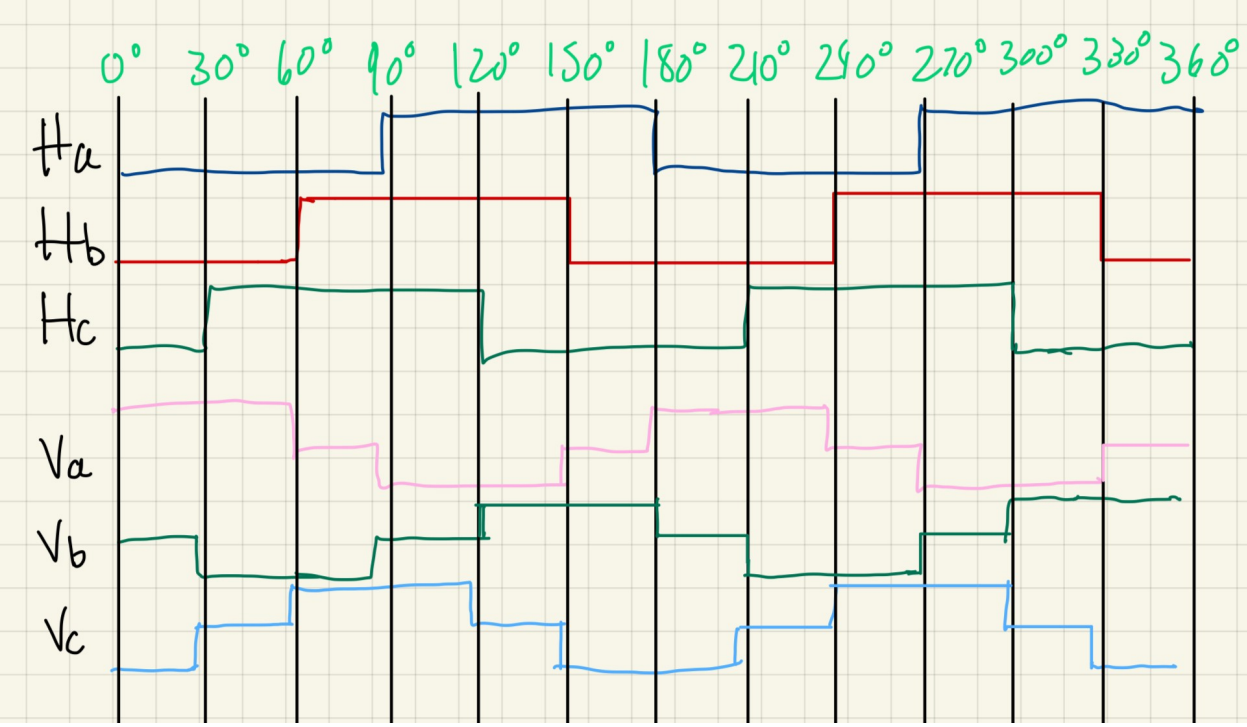
Permanent Brushless DC motors are one of the most commonly used motors in the world. Compared to a brushed motor, they require less maintenance, are more reliable, and produce good power. Learning the characteristics of them as an inductive load, and additionally how to drive them, is necessary and essential for us as students of computer engineering and embedded systems. Additionally, we will be learning the importance of hall effect sensors for brushless motor operation.

Initial Experiments

To start off, we stuck a plastic disc with angle measurements on the motor, then marking a point as 0 degrees. Afterwards, we manually energized the coils for a full rotation, and measured the coils. Each mechanical rotation of the motor was 60 degrees,with 6 full rotations to move 360 degrees. Afterwards, we plugged in and setup the hall effects sensors.

The hall effects sensors in relation to mecahnical movement and power polarity is shown below. In addition, a plot of the system on the oscilliscope is included. Using all of the information from above and the hall effects sensors, we began the next steps.

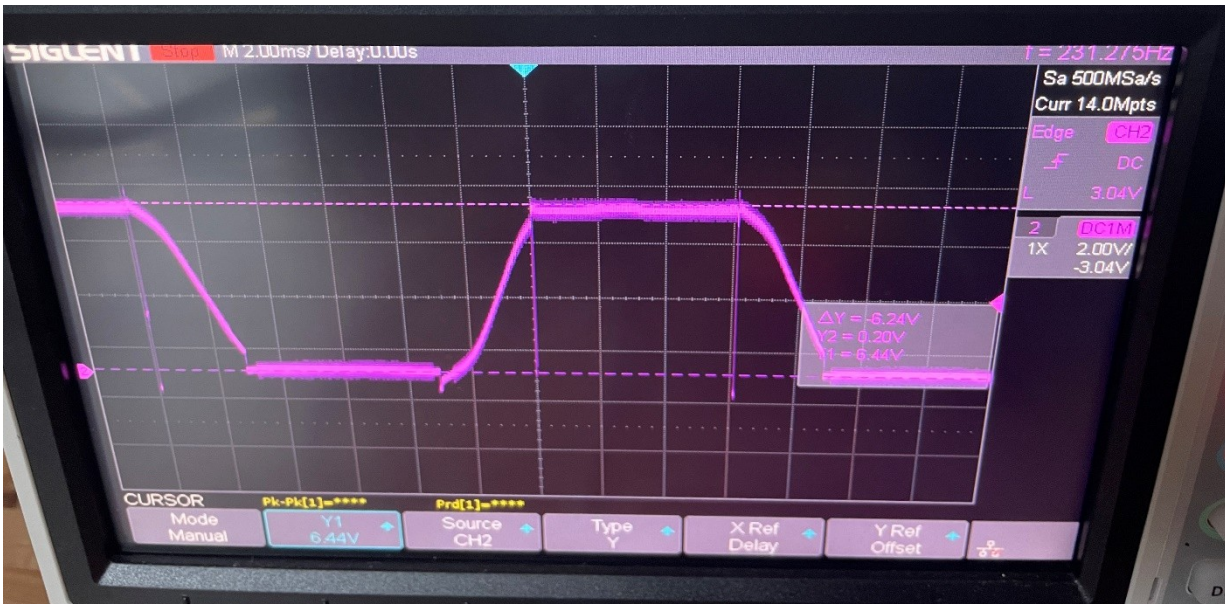




Further Experiments

After mapping out the hall effects sensors and their relation to motor phase, we created a program on a TM4C microcontroller to commutate the motor through its phases. We managed to get a steady and fairly power efficient rotation, with the lowest wait time between commutations being 2300ms. At this speed, the power usage was drastically lower and was close to around 2 watts and one amp. At higher wait times, the motor began to slip and run with much worse efficiency.

The final experiment we ran involved using the hall effects sensors to commutate the motor between its phases. Using a combination of a polling loop and some if statements, as well as a function that tracks the motor commutative state, we created a function that rotates the motor in phase with the hall effects sensors. After testing it by slowing down the motor, as well as noting vastly reduced power consumption, we looked at the back EMF from coil C. The back EMF was between 6 volts and 0 volts, and it looked like a trapezoidal shape. The figure is included below.



Conclusions

From our experiments, we discovered that Hall effect sensors are essential in brushless motors primarily because they can accurately detect rotor position. This capability is vital for the efficient operation of these motors, which depend on precise timing to switch the phases of the motor coils. Hall effect sensors deliver real-time feedback on the rotor's position, allowing the controller to activate the correct motor phases at the appropriate moments. Given these findings, brushless DC motors are relatively straightforward. They demonstrate a clear relationship between magnetic phases and voltages across several coils, and when you analyze the system, it becomes quite simple. Because of these advantages, it’s no surprise that they are widely used in everyday applications.

Circuit Built:

